

PiperABM: A Python Library for Resilience-Based Agent Modeling

Aslan Noorhasemi¹, Sierra Hicks², and Christopher McComb¹

¹ Department of Mechanical Engineering, Carnegie Mellon University, USA  ² Department of Natural Resources and the Environment, Cornell University, USA   Corresponding author

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

PiperABM is an open-source Python library for building agent-based simulations that couple human behavior with degrading infrastructure on real-world geospatial layouts. In *PiperABM*, infrastructure includes homes, markets, and the roads connecting them. The library provides lightweight primitives to enable agents with a customizable decision loop, infrastructure with spatial graphs whose effective travel costs evolve, and built-in analysis for accessibility to resources and travel distance, plus optional animation. Because it is implemented with familiar scientific tools like NumPy and NetworkX, researchers can inspect and extend models directly, integrate empirical data, and reuse standard workflows. *PiperABM* supports resilience research by examining how people's choices and infrastructure conditions co-evolve, seeking to reduce the setup overhead typical of one-off models while remaining flexible enough for diverse hazards, places, and policies.

Statement of Need

One established definition describes resilience as a community's capacity to withstand, adapt to, and recover from disruptions while maintaining or restoring essential functions ([United Nations Office for Disaster Risk Reduction, 2017](#)). *PiperABM* provides a flexible foundation for resilience research: users can adopt study-specific definitions and metrics, customize agent behavior and infrastructure dynamics, and analyze simulation outputs accordingly.

Agent-based models (ABMs) represent systems “from the bottom up” using heterogeneous, interacting agents whose local rules and behaviors generate macro-level patterns ([Joshua M. Epstein, 1999](#); [Joshua M. Epstein & Axtell, 1996](#); [Schelling, 2006](#)). ABM has also been highlighted as a framework that engages diverse stakeholders, enables the forecasting of equitable futures, and empowers policy ([Hicks et al., 2021](#); [Noorhasemi, Xiao, & McComb, 2025](#)). Over recent decades, ABM has matured into a practical research method supported by the scientific Python ecosystem ([Hoeven et al., 2025](#)). This paradigm is well-suited to infrastructure and community resilience, particularly within the Food–Energy–Water (FEW) nexus, where access to essential resources depends on both physical networks (e.g., roads linking homes to markets) and human decisions, and where nonlinear feedbacks, thresholds, and cascades are common ([Cansino-Loeza et al., 2022](#); [Noorhasemi, Xiao, Sha, et al., 2025](#); [Xue et al., 2024](#); [Zhang et al., 2019](#)).

Resilience studies pose several practical modeling needs that are not well served by ad hoc scripts or purely equation-based approaches:

- **Partial service levels and progressive degradation:** Models should represent graded performance (e.g., rougher roads increasing cost/time) rather than only binary failure, reflecting well-studied pavement deterioration processes and performance impacts, e.g., IRI-based assessments ([Altarabsheh et al., 2021](#); [Llopis-Castelló et al., 2024](#)).

- **Two-way coupling of behavior and infrastructure:** People adapt routes, priorities, and exchanges as networks evolve; these adaptations, in turn, load and degrade networks. Coupled human–infrastructure ABMs have demonstrated such feedbacks in flood-risk systems and related reviews emphasize capturing co-evolution ([Anshuka et al., 2022](#); [Michaelis et al., 2020](#)).
- **Real geospatial layouts for place-based analysis:** Integrating GIS data (maps/satellite-derived inputs) enables site-specific scenarios and stakeholder communication; recent work in the Python ABM ecosystem formalizes GIS ingestion and spatial operations for ABMs ([Wang et al., 2022](#)).
- **Ensemble experiments and reproducibility:** Parameter sweeps, stochastic replications, and uncertainty and sensitivity analysis require clear, standardized reporting and careful data handling; community guidelines for documenting agent-based models and recent work on calibration and sensitivity analysis support replicable workflows ([Grimm et al., 2020](#); [McCulloch et al., 2022](#); [Razavi et al., 2021](#)).
- **Custom metrics and analysis pipelines:** Familiar data structures expose model state for domain-specific metrics, such as spatiotemporal accessibility under disruption ([Enderami et al., 2024](#); [Tariverdi et al., 2023](#)).

PiperABM is built to address these needs and reduce the setup effort required for place-based resilience studies.

Intended users are researchers and practitioners in infrastructure and community resilience, including urban planners, policymakers, and decision-makers who need transparent, reproducible, place-based simulations.

Comparison to Existing Tools

We group related work into two domains: (1) General frameworks intended to be reusable across problems, and (2) Purpose-built models created for a single, specific study.

Domain 1: General frameworks (Python)

Mesa is a widely used general-purpose ABM framework that makes it easy to define agents, schedule steps, collect data, and visualize models in the browser, ideal for rapid prototyping and analysis ([Hoeven et al., 2025](#)). Trade-off: domain logic is do-it-yourself; researchers typically hand-roll infrastructure degradation, accessibility metrics, and behavior–infrastructure coupling. InfraRisk is an infrastructure-centric Python platform for interdependent networks (power, water, transport) with hazard, cascade, recovery, and resilience modules that is strong on network physics and restoration ([Balakrishnan & Cassottana, 2022](#)). Human decision-making and accessibility-driven behavior are typically added on top, so tight socio-technical feedbacks can be awkward. Repast4Py is a Python interface to an HPC engine for distributed/parallel ABMs; models can run across many cores or machines for large studies ([Collier & Ozik, 2022](#)). It does not provide resilience-specific logic (e.g., continuous degradation, accessibility metrics), which the modeler must supply.

Like Mesa and Repast4Py, PiperABM stays flexible and Pythonic, but it elevates resilience features to first-class concepts: continuous infrastructure degradation (e.g., rougher roads increasing travel cost), built-in accessibility/travel metrics, a lightweight OODA-style policy interface, geospatial ingestion, and optional animation. Unlike InfraRisk’s “network-first with agents layered on,” PiperABM starts from coupled behavior–infrastructure dynamics, making socio-technical feedbacks straightforward to model and analyze.

Domain 2: Purpose-built, one-off models

Examples include multilayer post-disaster recovery (e.g., Hurricane Harvey) ([Xue et al., 2024](#)), RecovUS for household recovery after Sandy ([Moradi & Nejat, 2020](#)), storm-induced power-

89 outage restoration with crew dispatch (Walsh et al., 2018), community resilience under tornado
90 hazards (Aghababaei & Koliou, 2023), and flood risk–insurance dynamics (Dubbelboer et
91 al., 2017). These models capture exactly what they need but are hard to reuse and often
92 re-implement similar scaffolding.

93 PiperABM offers the custom feel of bespoke models while reducing rework: degradation and
94 accessibility are built in; OODA-style decision logic is plug-and-play; infrastructure and agents
95 live on an inspectable NetworkX backend; and results can be analyzed with standard Python
96 tools. This makes it easier to adapt a single codebase to new hazards, geographies, and
97 policies, and to compare scenarios consistently.

98 Model Overview

99 PiperABM couples infrastructure and society networks represented as NetworkX graphs, with
100 their interactions summarized in Figure 1.

101 The infrastructure network represents the physical layout: nodes represent homes, markets,
102 and junctions; edges represent streets and access connections from homes and markets to
103 streets. Spatial layouts can be constructed manually or imported from map data.

104 Society nodes represent individual agents, with family, neighbor, and friendship ties as edges.
105 Family ties connect agents assigned to the same home. Agents have a home assignment,
106 current location, monetary balance, socioeconomic status, resource inventories, and sufficiency
107 thresholds, with food, water, and energy as the default resources. Home assignments and
108 agent locations link the two networks. Markets hold resource inventories, while social ties
109 identify friends' and neighbors' homes as potential destinations.

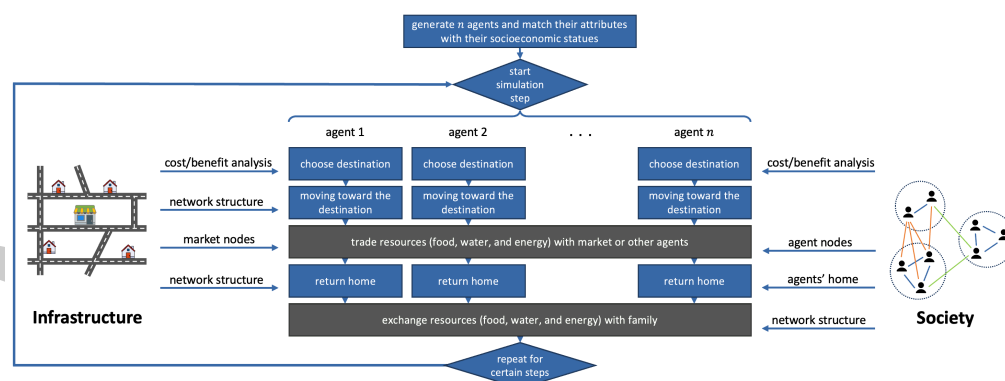


Figure 1: PiperABM overview: Infrastructure locations and social relationships shape agents' travel and resource exchanges.

110 The baseline decision mechanism follows OODA-style logic: observing, orienting, deciding, and
111 acting (Brehmer, 2005; Johnson, 2023). When an agent's queued actions finish, it selects a
112 new destination, considering markets first and socially connected homes if no suitable market
113 is found. Destinations are ranked by resource value minus estimated travel resource costs, with
114 travel-time constraints determining feasibility. Agents queue travel, a stay at the destination,
115 a return home, and rest. At each simulation step, the model coordinates resource exchanges
116 among agents at the same home or with markets where agents are present, using a cooperative,
117 centralized Nash bargaining formulation (Nash, 1950). Market inventories are replenished each
118 step.

119 Agents consume resources while staying and traveling, and travel accumulates usage
120 on infrastructure edges. With degradation enabled, usage and aging increase streets'

effective lengths, which determine A* routing costs (Foead et al., 2021) and travel duration. Infrastructure conditions therefore affect subsequent routes and feasible destinations, while agent movement contributes to further degradation.

Outputs include travel distance and resource accessibility, measured as inventory relative to a sufficiency threshold and capped at one. KeepDelta can store human-readable changes in model state (Noorghasemi & McComb, 2025), and optional animation supports face-validity assessment. Researchers can replace the baseline decision and degradation rules through user-defined extensions.

Acknowledgements

This work was supported by the U.S. National Science Foundation through Grant RISE-1927718. Any opinions, findings, conclusions, or recommendations expressed in this article are those of the authors and do not necessarily reflect the views of the sponsors.

References

- Aghababaei, M., & Koliou, M. (2023). Community resilience assessment via agent-based modeling approach. *Computer-Aided Civil and Infrastructure Engineering*, 38(7), 920–939. <https://doi.org/https://doi.org/10.1111/mice.12916>
- Altarabsheh, A., Altarabsheh, R., Altarabsheh, S., & Asi, I. (2021). Prediction of pavement performance using multistate survival models. *Journal of Transportation Engineering, Part B: Pavements*, 147(1), 04020082. <https://doi.org/10.1061/JPEODX.0000241>
- Anshuka, A., Ogtrop, F. F. van, Sanderson, D., & Leao, S. Z. (2022). A systematic review of agent-based model for flood risk management and assessment using the ODD protocol. *Natural Hazards*, 112(3), 2739–2771. <https://doi.org/10.1007/s11069-022-05286-y>
- Balakrishnan, S., & Cassottana, B. (2022). InfraRisk: An open-source simulation platform for resilience analysis in interconnected power–water–transport networks. *Sustainable Cities and Society*, 83, 103963. <https://doi.org/https://doi.org/10.1016/j.scs.2022.103963>
- Brehmer, B. (2005). The dynamic OODA loop: Amalgamating boyd's OODA loop and the cybernetic approach to command and control. *Proceedings of the 10th International Command and Control Research Technology Symposium*, 365–368.
- Cansino-Loeza, B., Munguía-López, A. del C., & Ponce-Ortega, J. M. (2022). A water-energy-food security nexus framework based on optimal resource allocation. *Environmental Science & Policy*, 133, 1–16. <https://doi.org/10.1016/j.envsci.2022.03.006>
- Collier, N., & Ozik, J. (2022). Distributed agent-based simulation with Repast4Py. 2022 *Winter Simulation Conference (WSC)*, 192–206. <https://doi.org/10.1109/WSC57314.2022.10015389>
- Dubbelboer, J., Nikolic, I., Jenkins, K., & Hall, J. (2017). An agent-based model of flood risk and insurance. *Journal of Artificial Societies and Social Simulation*, 20(1), 6. <https://doi.org/10.18564/jasss.3135>
- Enderami, S. A., Sutley, E., Helgeson, J., Dueñas-Osorio, L., Watson, M., & Lindt, J. W. van de. (2024). Measuring post-disaster accessibility to essential goods and services: Proximity, availability, adequacy, and acceptability dimensions. *Journal of Infrastructure Preservation and Resilience*, 5(1). <https://doi.org/10.1186/s43065-024-00104-0>
- Epstein, Joshua M. (1999). Agent-based computational models and generative social science. *Complexity*, 4(5), 41–60.
- Epstein, Joshua M., & Axtell, R. (1996). *Growing artificial societies: Social science from the*

- bottom up. Brookings Institution Press; MIT Press. ISBN: 9780262550253
- Foead, D., Ghifari, A., Kusuma, M. B., Hanafiah, N., & Gunawan, E. (2021). A Systematic Literature Review of A* Pathfinding. *Procedia Computer Science*, 179, 507–514. <https://doi.org/10.1016/j.procs.2021.01.034>
- Grimm, V., Railsback, S. F., Vincenot, C. E., Berger, U., Gallagher, C., DeAngelis, D. L., Edmonds, B., Ge, J., Giske, J., Groeneveld, J., Johnston, A. S. A., Milles, A., Nabe-Nielsen, J., Polhill, J. G., Radchuk, V., Rohwäder, M.-S., Stillman, R. A., Thiele, J. C., & Ayllón, D. (2020). The ODD protocol for describing agent-based and other simulation models: A second update to improve clarity, replication, and structural realism. *Journal of Artificial Societies and Social Simulation*, 23(2), 7. <https://doi.org/10.18564/jasss.4259>
- Hicks, S., McComb, C., & Alessa, L. (2021). *Agents and the Arctic: The Case for Increased Use of Agent-Based Modeling to Study Permafrost* — [thearcticinstitute.org/agents-arctic-case-increased-use-agent-based-modeling-study-permafrost/](https://www.thearcticinstitute.org/agents-arctic-case-increased-use-agent-based-modeling-study-permafrost/).
- Hoeven, E. ter, Kwakkel, J., Hess, V., Pike, T., Wang, B., rht, & Kazil, J. (2025). Mesa 3: Agent-based modeling with Python in 2025. *Journal of Open Source Software*, 10(107), 7668. <https://doi.org/10.21105/joss.07668>
- Johnson, J. (2023). Automating the OODA loop in the age of intelligent machines: Reaffirming the role of humans in command-and-control decision-making in the digital age. *Defence Studies*, 23(1), 43–67. <https://doi.org/10.1080/14702436.2022.2102486>
- Llopis-Castelló, D., Camacho-Torregrosa, F. J., Romeral-Pérez, F., & Tomás-Martínez, P. (2024). Estimation of pavement condition based on data from connected and autonomous vehicles. *Infrastructures*, 9(10). <https://doi.org/10.3390/infrastructures9100188>
- McCulloch, J., Ge, J., Ward, J., Heppenstall, A., Polhill, J. G., & Malleson, N. (2022). Calibrating agent-based models using uncertainty quantification methods. *Journal of Artificial Societies and Social Simulation*, 25(2), 1. <https://doi.org/10.18564/jasss.4791>
- Michaelis, T., Brandimarte, L., & Mazzoleni, M. (2020). Capturing flood-risk dynamics with a coupled agent-based and hydraulic modelling framework. *Hydrological Sciences Journal*, 65(9), 1458–1473. <https://doi.org/10.1080/02626667.2020.1750617>
- Moradi, S., & Nejat, A. (2020). RecovUS: An agent-based model of post-disaster household recovery. *Journal of Artificial Societies and Social Simulation*, 23(4), 13. <https://doi.org/10.18564/jasss.4445>
- Nash, J. F. (1950). The Bargaining Problem. *Econometrica*, 18(2), 155–162. <https://doi.org/10.2307/1907266>
- Noorghasemi, A., & McComb, C. (2025). KeepDelta: A Python Library for Human-Readable Data Differencing. *Journal of Open Source Software*, 10(110), 8075. <https://doi.org/10.21105/joss.08075>
- Noorghasemi, A., Xiao, M., & McComb, C. (2025). *Evaluating community well-being and resilience via agent-based models*. <https://doi.org/10.22541/essoar.173687385.52831340/v1>
- Noorghasemi, A., Xiao, Y., Sha, Z., Xiao, M., & McComb, C. (2025). Agent-based modeling for the evaluation of community resilience in silico. *Authorea Inc.*
- Razavi, S., Jakeman, A., Saltelli, A., Prieur, C., Iooss, B., Borgonovo, E., Plischke, E., Lo Piano, S., Iwanaga, T., Becker, W., Tarantola, S., Guillaume, J. H. A., Jakeman, J., Gupta, H., Melillo, N., Rabitti, G., Chabridon, V., Duan, Q., Sun, X., ... Maier, H. R. (2021). The future of sensitivity analysis: An essential discipline for systems modeling and policy support. *Environmental Modelling & Software*, 137, 104954. <https://doi.org/https://doi.org/10.1016/j.envsoft.2020.104954>

- 213 Schelling, T. C. (2006). *Micromotives and macrobehavior* (Revised). W. W. Norton &
214 Company. <https://doi.org/10.2307/134812>
- 215 Tariverdi, M., Nunez-del-Prado, M., Leonova, N., & Rentschler, J. (2023). Measuring
216 accessibility to public services and infrastructure criticality for disasters risk management.
217 *Scientific Reports*, 13(1). <https://doi.org/10.1038/s41598-023-28460-z>
- 218 United Nations Office for Disaster Risk Reduction. (2017). *The Sendai Framework Terminology*
219 *on Disaster Risk Reduction: Resilience*. <https://www.undrr.org/terminology/resilience>
- 220 Walsh, T., Layton, T., Wanik, D., & Mellor, J. (2018). Agent based model to estimate time
221 to restoration of storm-induced power outages. *Infrastructures*, 3(3). <https://doi.org/10.3390/infrastructures3030033>
- 222
- 223 Wang, B., Hess, V., & Crooks, A. (2022). Mesa-Geo: A GIS Extension for the Mesa
224 Agent-Based Modeling Framework in Python. *Proceedings of the 5th ACM SIGSPATIAL*
225 *International Workshop on GeoSpatial Simulation*, 1–10. <https://doi.org/10.1145/3557989.3566157>
- 226
- 227 Xue, J., Park, S., Mondal, W. U., Reia, S. M., Yao, T., & Ukkusuri, S. V. (2024). An agent-
228 based model of post-disaster recovery in multilayer socio-physical networks. *Sustainable*
229 *Cities and Society*, 115, 105863. <https://doi.org/https://doi.org/10.1016/j.scs.2024.105863>
- 230
- 231 Zhang, P., Zhang, L., Chang, Y., Xu, M., Hao, Y., Liang, S., Liu, G., Yang, Z., & Wang,
232 C. (2019). Food-energy-water (FEW) nexus for urban sustainability: A comprehensive
233 review. *Resources, Conservation and Recycling*, 142, 215–224. <https://doi.org/https://doi.org/10.1016/j.resconrec.2018.11.018>
- 234

DRAFT